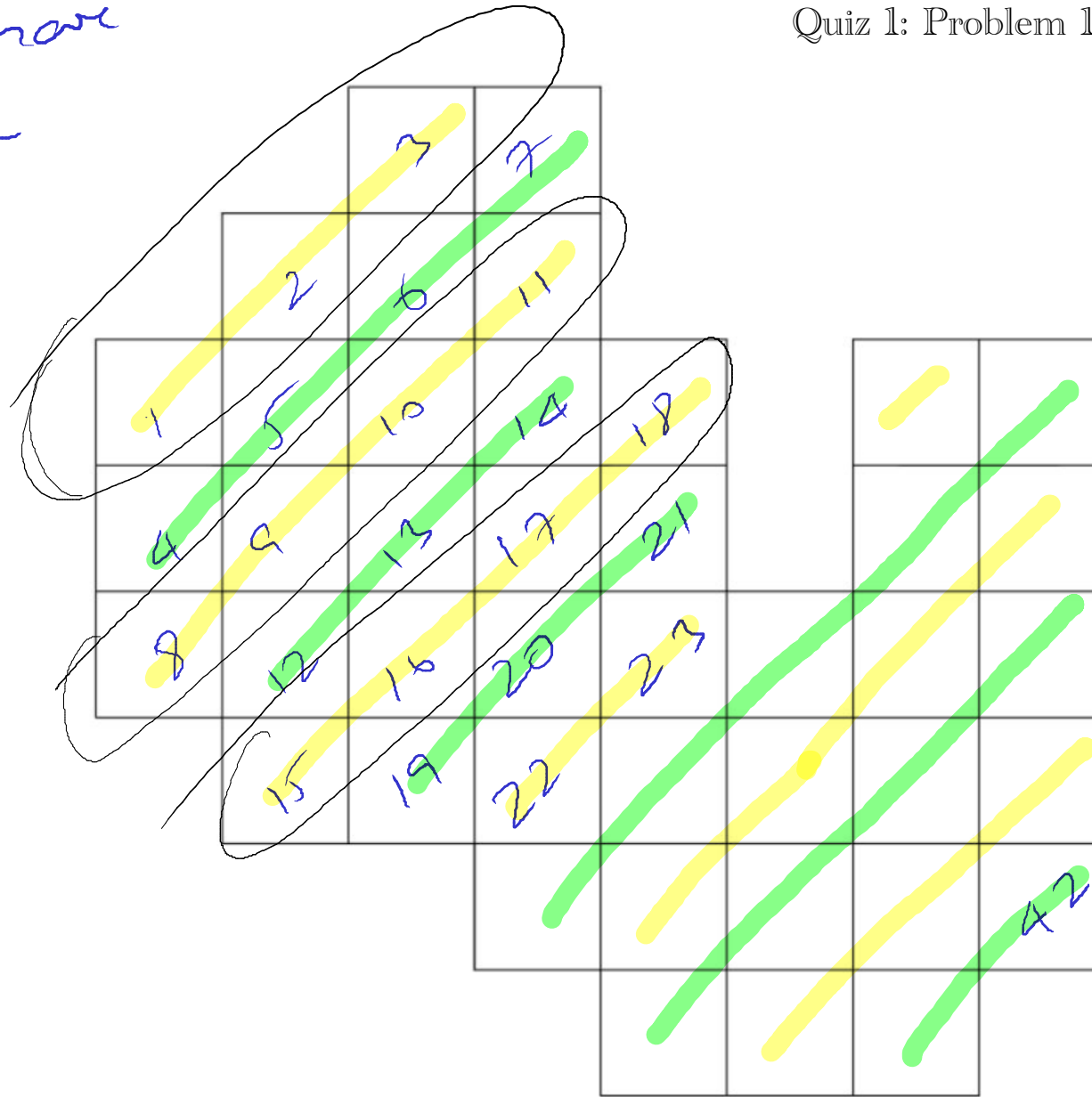


Edges: if they share
a side

Post matching
in graph
iff

holding

Quiz 1: Problem 1



Data Compression Problem

Goal! Minimize # of bits stored.

$$\Sigma = \{A, B, C, D\}$$

$$s \in \Sigma^{\text{length}}$$

$$\# \text{ bits} = \sum_{i=1}^4 f_i \cdot 2 = 2|s|$$

Idea! Use shorter code word

for most freq and
longer for less freq.

Code Book	
A	00
B	01
C	10
D	11

0 0 0 1 0	0 0 0 1 1 0 1 1 0	A	60	0	0
A A A C	A A B C D A	B	20	0 1	1 0
B A C	A A A D B C	C	15	1 0	1 1 0
A B C		D	5	1 1	1 1 1

Idea: If no codeword is prefix of another then we can uniquely decode.

Prefix free codes

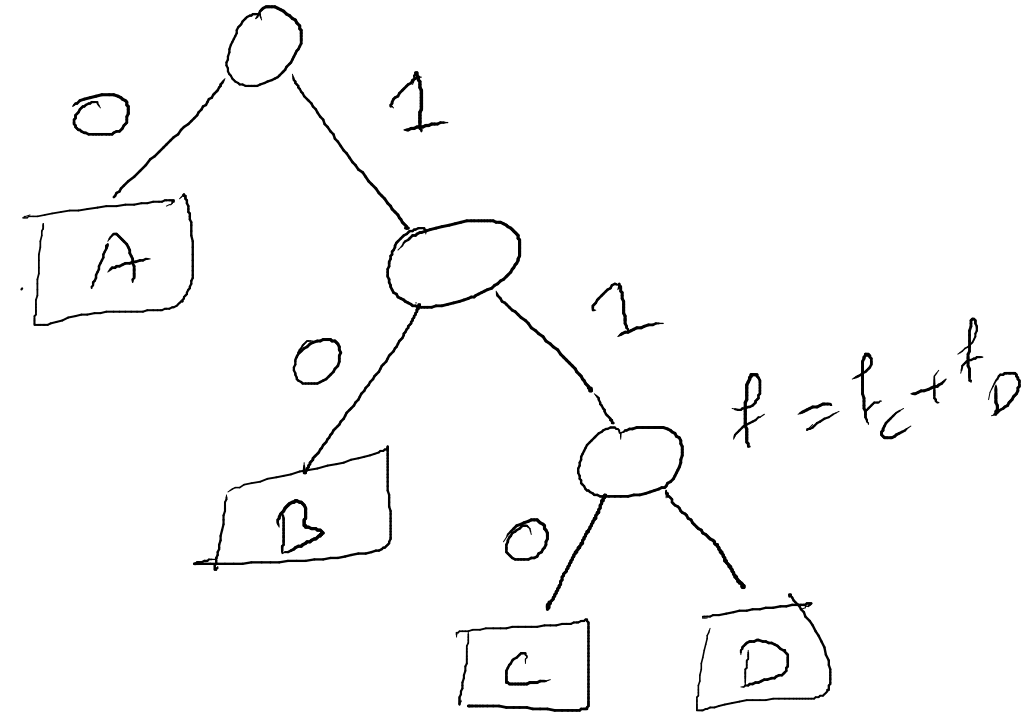
Idea: A prefix free code can be drawn as a binary tree

Every node has deg 0 or deg 2

Leaf nodes corresponds to alphabets.

Goal! $\min \sum_{i=1}^4 f_i \times l_i = f_A \cdot 1 + f_B \cdot 2 + f_C \cdot 3 + f_D \cdot 3$

$\nearrow$ depⁿ or leaf.

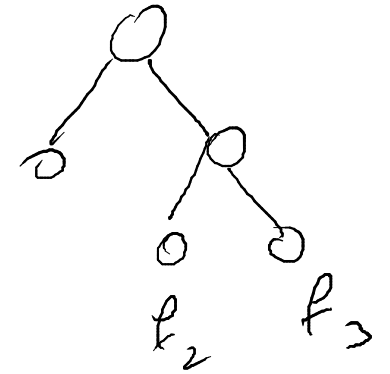


Design Alg to find best tree that minimize size

$$\text{Size} = \sum_{i=1}^n f_i l_i = \sum_{i=1}^n f_i + \sum_{i \in \text{Internal} \setminus \text{Root}} f_i$$

$$f_1 + 2f_2 + 2f_3$$

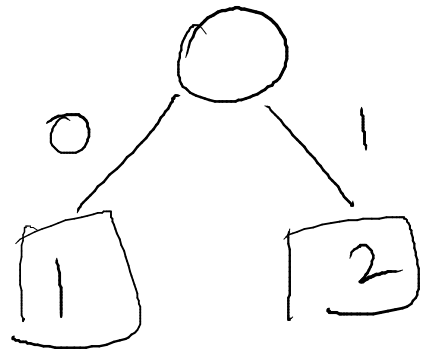
$$= f_1 + f_2 + f_3 + (f_2 + f_3)$$



$$f_1 < f_2 < f_3$$

$< f_3$. Then f_1, f_2 need to be siblings and has the higher f_3 as a sibling.

f_1 and f_2 should have the highest depth
and they can be siblings.



3

4

...

97

.

1

$$f_1 = f_1 + f_2$$

3

4

f_3

f_4

97

f_{97}

Read: Section 5.2, in Dasgupta, Papadimiriu, Vazirani

Solve

1. Show that every d regular bipartite graph (all vertices on the left and right has degree $d > 0$), has a perfect matching.
2. Show that every d regular bipartite graph is the union of d perfect matchings.
3. Show that for a prefix free code,

$$\sum_{i=1}^m 2^{-l_i} \leq 1$$

- $\sum$: The summation symbol, adding values from $i = 1$ to m .
- m : The total number of codewords in the codebook.
- l_i : The length (number of symbols/bits) of the i -th codeword.